

15. *Assertion (A) :* Electrostatic force is a conservative force.

Reason (R) : In an electrostatic field, the work done between two points per unit positive charge depends on the path followed.

CBSE 25 C

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Reason (R) : In an electrostatic field, the work done between two points per unit positive charge depends on the path followed.

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(C) Assertion (A) is true, but reason (R) is false

- (b) (i) An electric dipole of dipole moment p consists of point charges q and $-q$, separated by $2a$. Derive an expression for electric potential in terms of its dipole moment at a point at a distance x ($\gg a$) from its centre and lying (I) along its axis, and (II) along its bisector line.

I. Along its axis

$$V_- = \frac{-kq}{x+a}$$

$$V_+ = \frac{kq}{x-a}$$

$$V = V_- + V_+$$

$$= kq \left(\frac{-1}{x+a} + \frac{1}{x-a} \right)$$

$$= kq \frac{2a}{(x^2 - a^2)} = \frac{kp}{x^2 - a^2}$$

$$x \gg a \quad \therefore V = \frac{kp}{x^2}$$

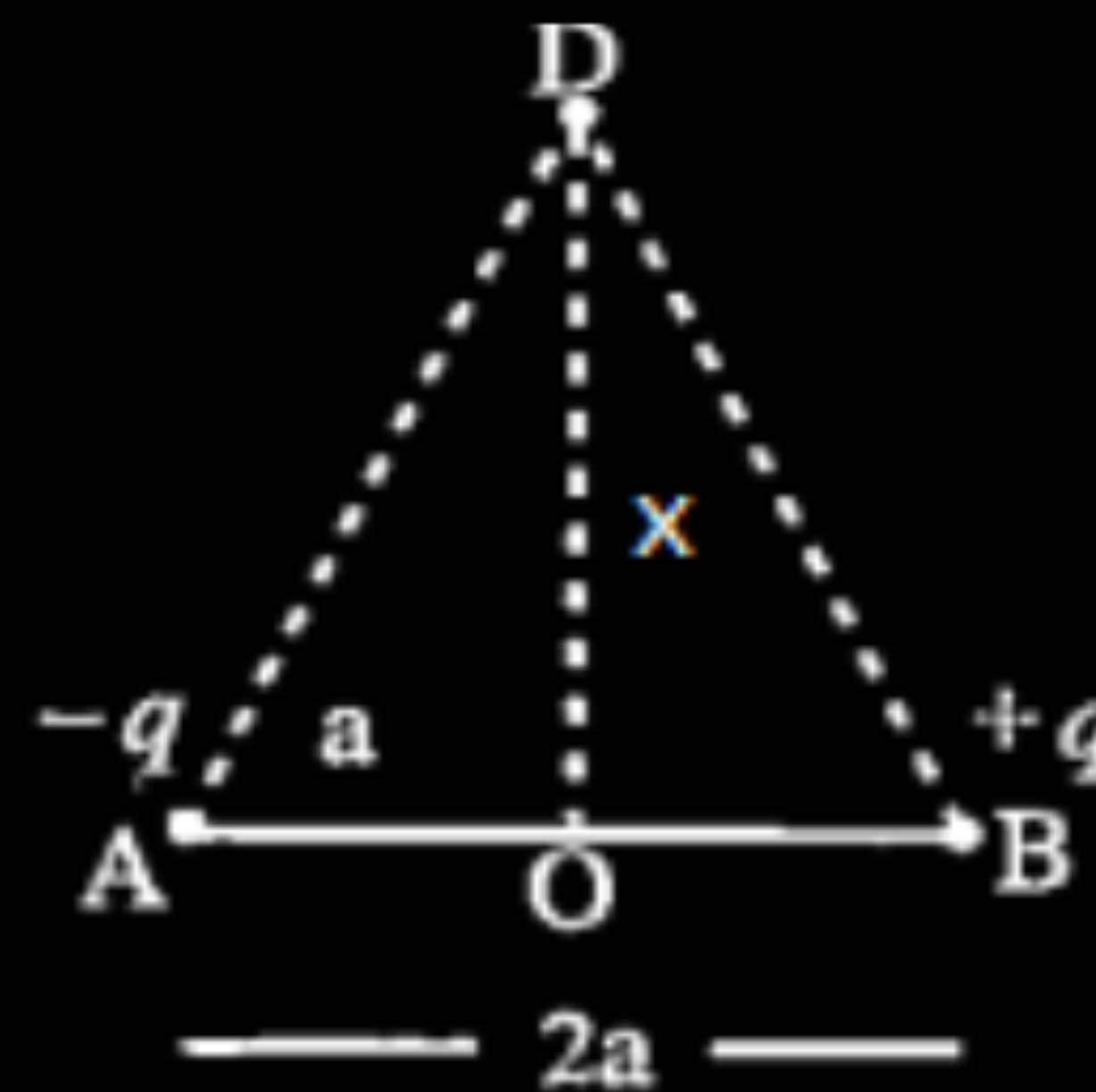
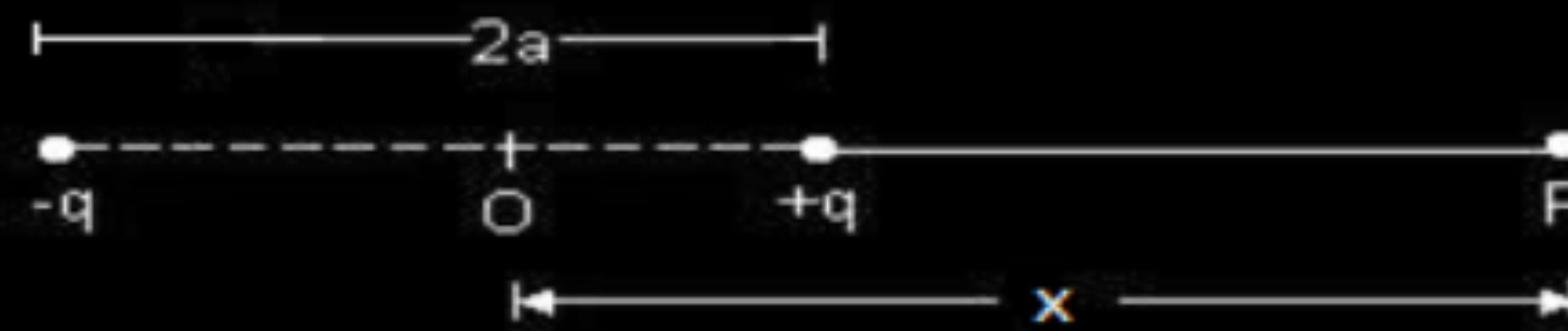
II. Along the bisector line

$$V_- = \frac{kq}{\sqrt{x^2 + a^2}}$$

$$V_+ = \frac{-kq}{\sqrt{x^2 + a^2}}$$

$$V = V_- + V_+$$

$$= 0$$



1/2

1/2

1/2

1/2

1/2

1/2

1. The electric field (E) and electric potential (V) at a point inside a charged hollow metallic sphere are respectively :

(A) $E = 0, V = 0$

(B) $E = 0, V = V_0$ (a constant)

(C) $E \neq 0, V \neq 0$

(D) $E = E_0$ (a constant), $V = 0$

1. The electric field (E) and electric potential (V) at a point inside a charged hollow metallic sphere are respectively :
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 - (B) $E = 0, V = V_0$ (a constant)
 - (C) $E \neq 0, V \neq 0$
 - (D) $E = E_0$ (a constant), $V = 0$

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(B) $E = 0, V = V_0$ (a constant)

- 31.** (a) (i) The electric field in a region is given by $\vec{E} = 40x \hat{i}$ N/C. Find the amount of work done in taking a unit positive charge from a point (0, 3m) to the point (5m, 0).
- (ii) A charge Q is distributed over two concentric hollow spheres of radii r and R (> r) such that their surface charge densities are equal. Find :
- (I) the electric field, and
- (II) the potential
- at their common centre.

(a)

$$\begin{aligned} \text{(i)} \quad V &= - \int \vec{E} \cdot d\vec{r} \\ &= - \int 40x dx \\ &= -20x^2 \end{aligned}$$

Potential at A (0, 3m), $V_A = 0$

Potential at B (5m, 0), $V_B = -500 \text{ V}$

Work done in taking a unit positive charge from a point (0, 3m) to the point (5m, 0)

$$\begin{aligned} W &= q(V_B - V_A) \\ &= 1(-500 - 0) \end{aligned}$$

$$W = -500 \text{ J}$$

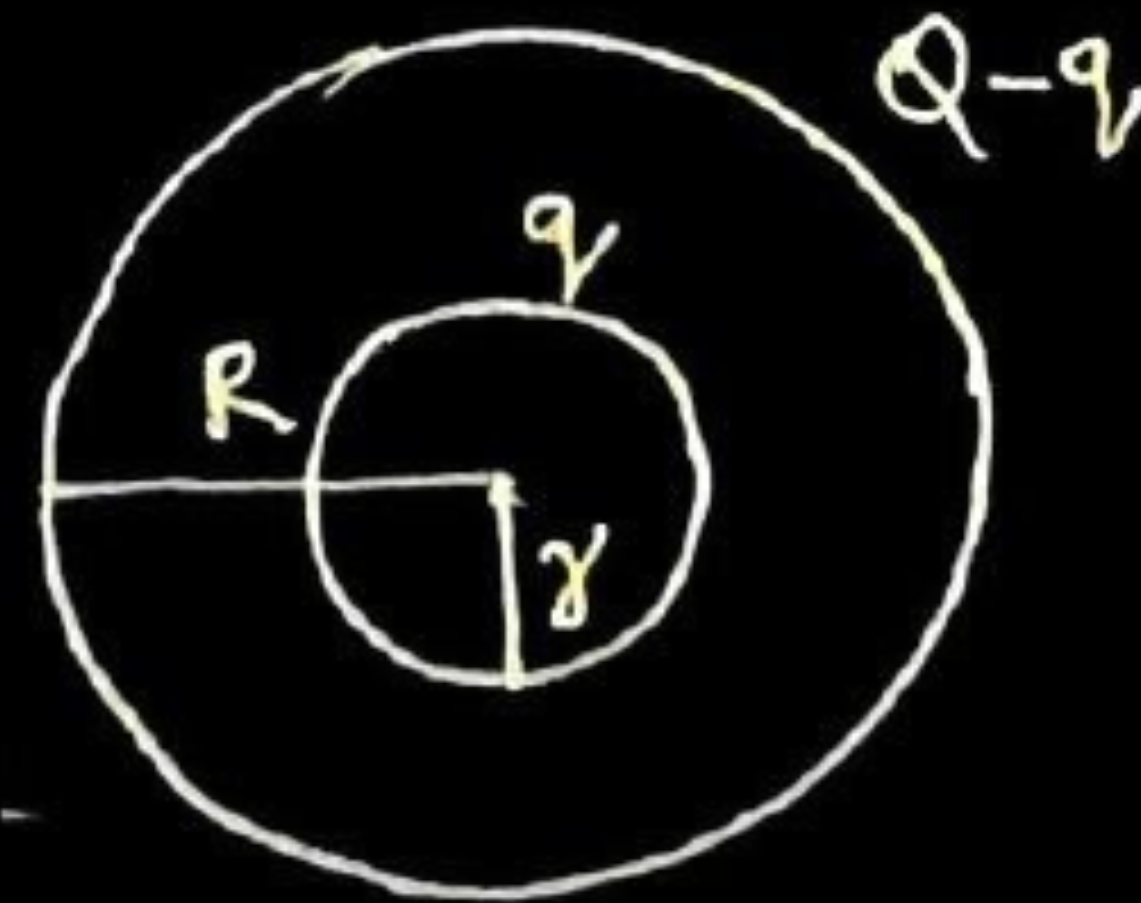
(ii) (I) Electric field at the common centre will be zero as the charge enclosed by the inner sphere is zero.

Alternatively: $q_{en} = 0$

$$\phi_E = 0$$

$$\oint \vec{E} \cdot d\vec{s} = 0$$

$$E = 0$$



1/2

1/2

1/2

1/2

1

(II) $\therefore$ Surface charge densities are equal

$$\begin{aligned} \frac{q}{4\pi r^2} &= \frac{Q-q}{4\pi R^2} \\ q &= \frac{Qr^2}{R^2 + r^2} \end{aligned}$$

Potential at common centre

$$V = \frac{kq}{r} + \frac{k(Q-q)}{R}$$

$$V = \frac{k}{r} \frac{Qr^2}{(R^2 + r^2)} + \frac{k}{R} \left[Q - \frac{Qr^2}{(R^2 + r^2)} \right]$$

$$V = \frac{kQ(R+r)}{R^2 + r^2}$$

- 26.** (a) Two small solid metal balls A and B of radii R and $2R$ having charge densities 2σ and 3σ respectively are kept far apart. Find the charge densities on A and B after they are connected by a conducting wire.

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For ball A

$$q_1 = 2\sigma \times 4\pi R^2$$

$$= 8\pi R^2 \sigma$$

For ball B

$$q_2 = 3\sigma \times 4\pi (2R)^2$$

$$= 48\pi R^2 \sigma$$

$$\text{Total charge (Q)} = q_1 + q_2$$

$$= 56\pi R^2 \sigma$$

When balls A and B are connected by a wire, their potentials will be equal
Let q be the charge on ball A and $(Q - q)$ be the charge on the ball B after connecting wire.

$$\frac{Kq}{R} = \frac{K(Q - q)}{2R}$$

$$2q = Q - q$$

$$q = \frac{Q}{3}$$

$$= \frac{56\pi R^2 \sigma}{3}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$$Q - \frac{Q}{3} = \frac{112\pi R^2 \sigma}{3}$$

$$\sigma_A = \frac{\frac{56\pi R^2 \sigma}{3}}{4\pi R^2}$$

$$= \frac{14}{3} \sigma$$

$$\sigma_B = \frac{\frac{112\pi R^2 \sigma}{3}}{4\pi (2R)^2}$$

$$= \frac{7}{3} \sigma$$

$\frac{1}{2}$

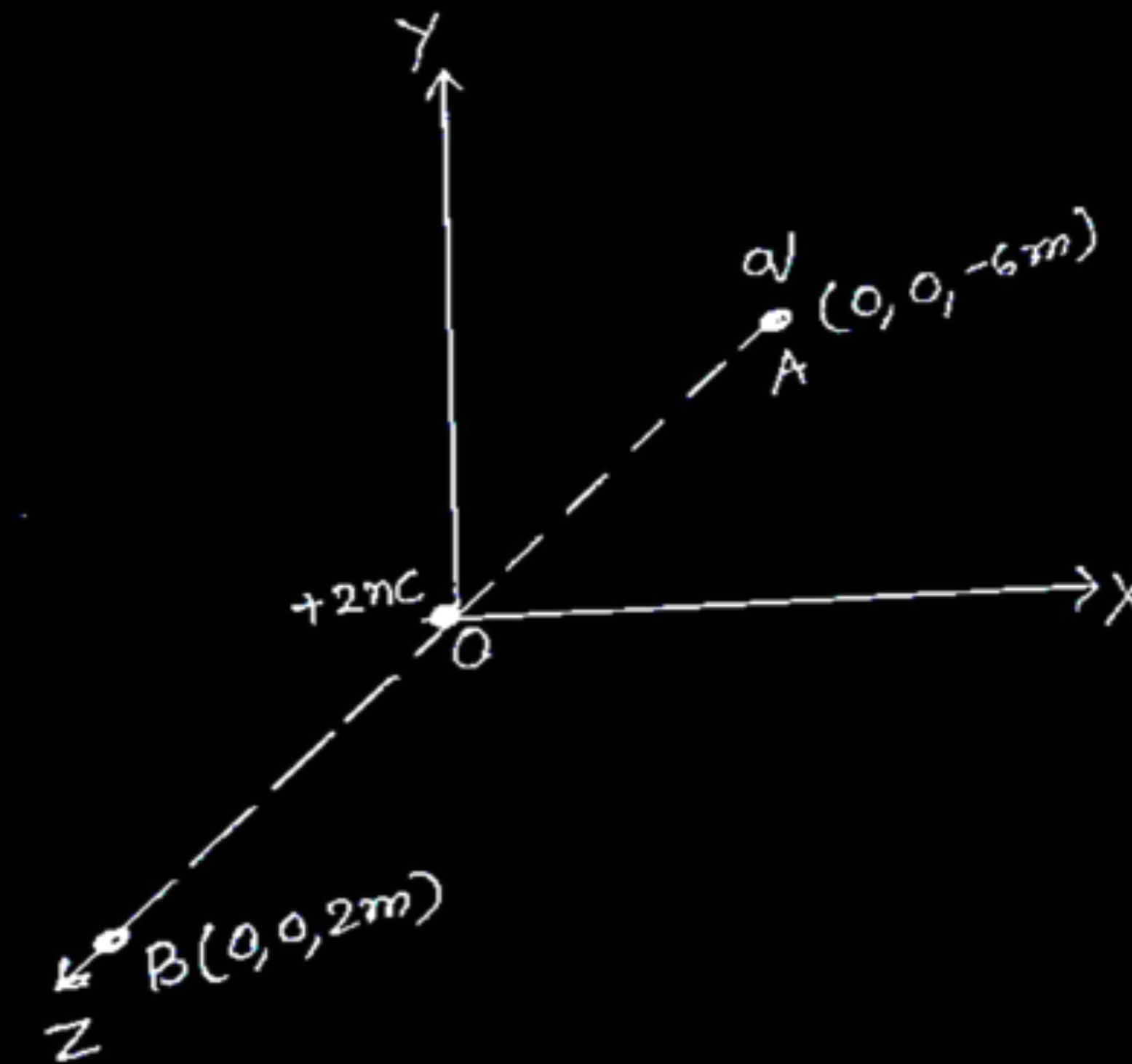
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- (ii) A point charge of $+2\text{ nC}$ is kept at the origin of a three-dimensional coordinate system. Find the type and magnitude of the charge which should be kept at $(0, 0, -6\text{ m})$ so that the potential due to the system becomes zero at $(0, 0, 2\text{ m})$.

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(ii)



Let the charge is kept at A be q

Potential at point B due to charge at the origin O and charge (q) at A

$$V = V_1 + V_2$$

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{2 \times 10^{-9}}{2} + \frac{q}{6+2} \right]$$

$$\frac{1}{4\pi\epsilon_0} \left[10^{-9} + \frac{q}{8} \right] = 0$$

$$q = -8 \times 10^{-9} \text{ C}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

3. Which one of the following statements is correct ?

Electric field due to static charges is

1

- (A) conservative and field lines do not form closed loops.
- (B) conservative and field lines form closed loops.
- (C) non-conservative and field lines do not form closed loops.
- (D) non-conservative and field lines form closed loops.

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3. Which one of the following statements is correct ?

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(A) conservative and field lines do not form closed loops.

33. (a) (i) A small conducting sphere A of radius r charged to a potential V , is enclosed by a spherical conducting shell B of radius R . If A and B are connected by a thin wire, calculate the final potential on sphere A and shell B.
- (ii) Write two characteristics of equipotential surfaces. A uniform electric field of 50 NC^{-1} is set up in a region along $+x$ axis. If the potential at the origin $(0, 0)$ is 220 V , find the potential at a point $(4\text{m}, 3\text{m})$.

5

(i) Potential on sphere A = $V = \frac{Q}{4\pi\epsilon_0 r}$

Charge on sphere A = $4\pi\epsilon_0 r V$

The charge is transferred to shell B.

Potential on shell B = $\frac{1}{4\pi\epsilon_0} \times \frac{4\pi\epsilon_0 r V}{R}$

Potential on shell B = $\frac{rV}{R}$

Potential on sphere A = Potential on shell B

(ii) Characteristics of equipotential surfaces: -
(Any two)

- Potential at all points on the surface is same.
- Equipotential surface is normal to the direction of the electric field.
- The work done in moving a charge on an equipotential surface is zero.

$V_0 - V = E d = 50 \times 4$

$V_0 - V = 200 \text{ V}$

$V = 220 \text{ V} - 200 \text{ V}$

$V = 20 \text{ V}$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2}$

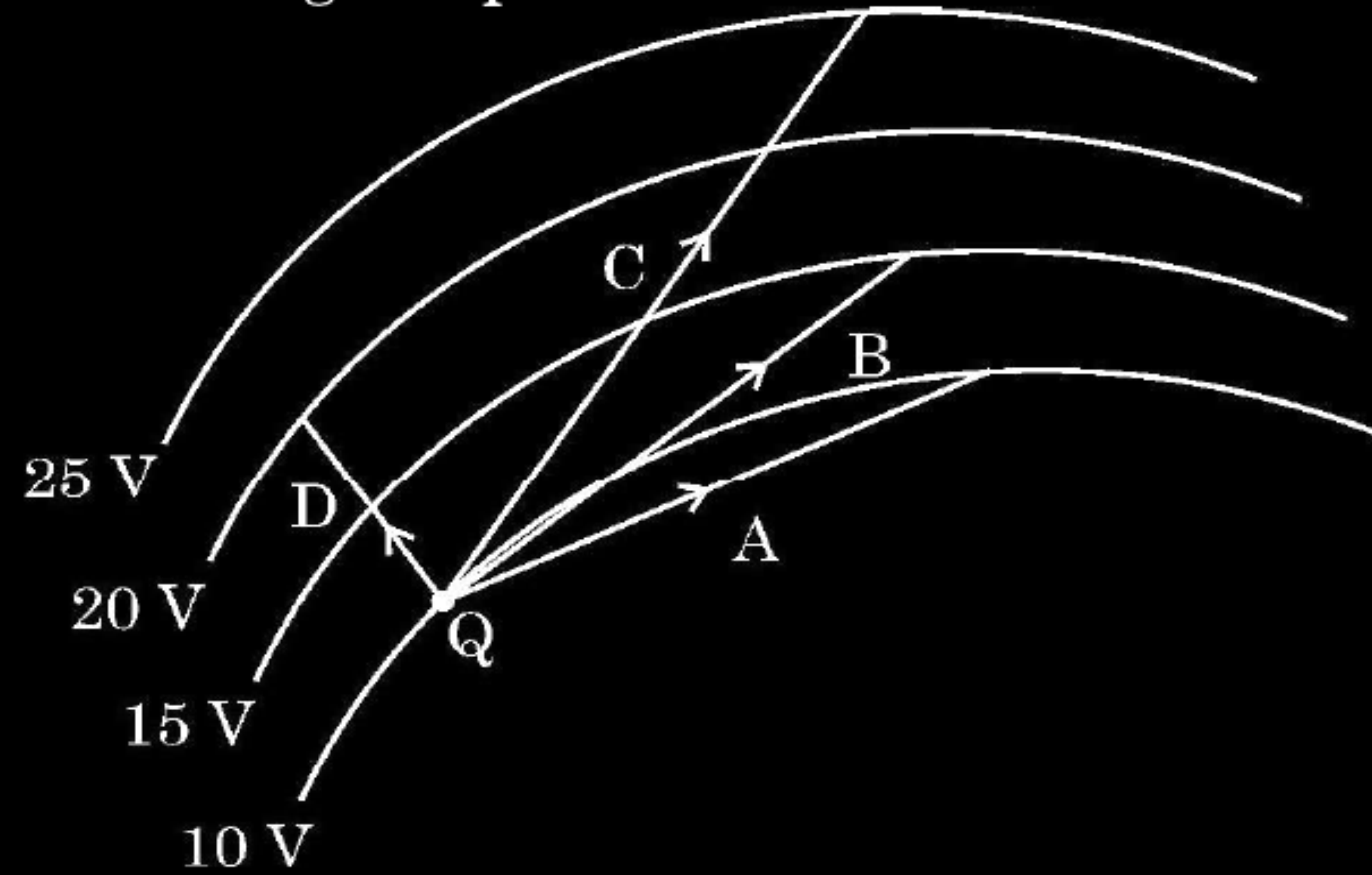
$\frac{1}{2}$

$\frac{1}{2}$

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1. In the figure curved lines represent equipotential surfaces. A charge Q is moved along different paths A, B, C and D. The work done on the charge will be maximum along the path

1



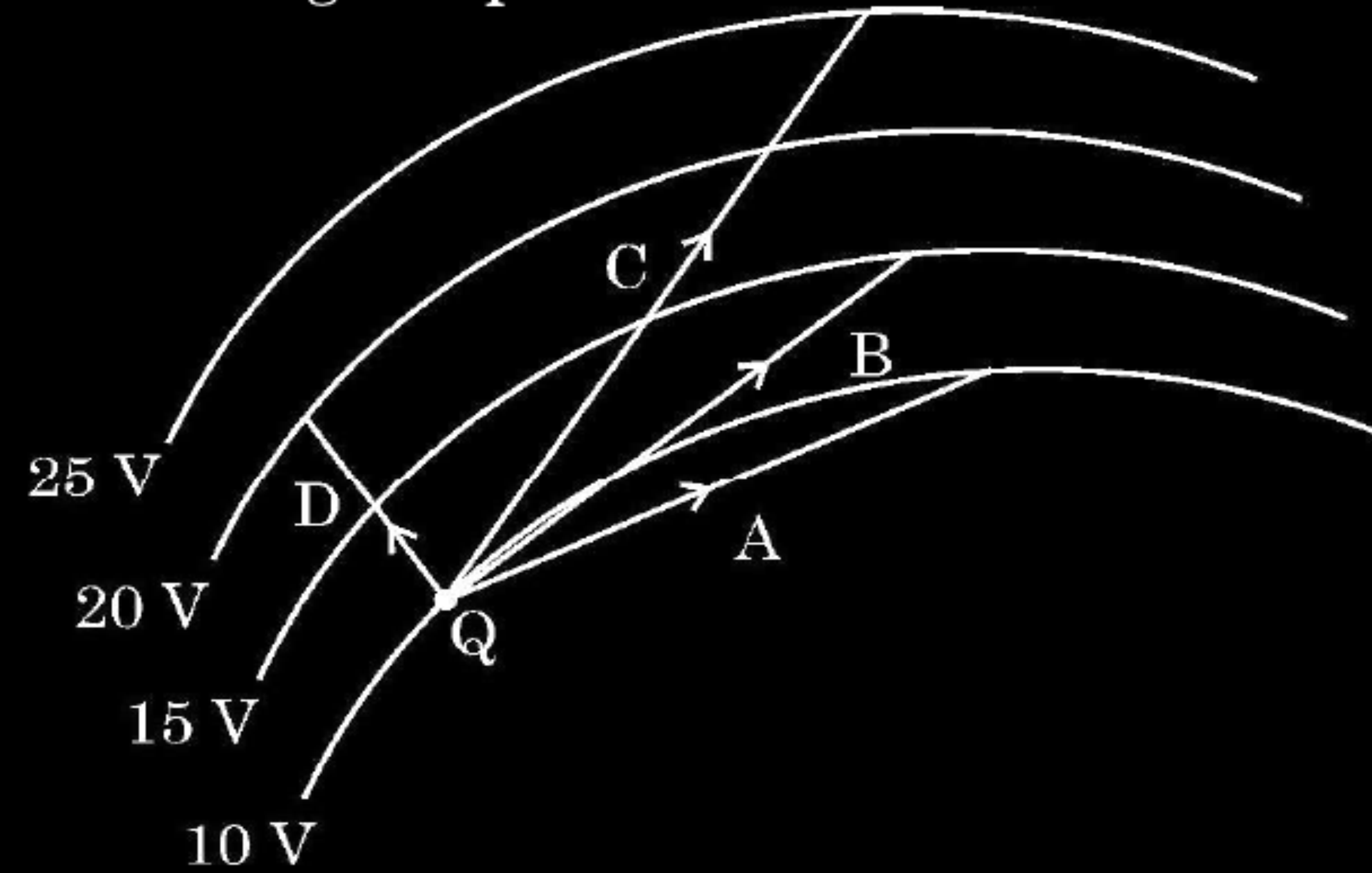
(A) A
(C) C

(B) B
(D) D

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(A) A
(C) C

(B) B
(D) D

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(C) C

3. A conducting sphere of radius R is given a charge Q . Consider three points A, B and C — A at the centre, B at a distance $R/2$ from the centre and C on the surface of the sphere. The electric potentials at these points are such that :

(A) $V_A = V_B = V_C$

(B) $V_A = V_B \neq V_C$

(C) $V_A \neq V_B \neq V_C$

(D) $V_A \neq V_B = V_C$

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(D) $V_A \neq V_B = V_C$

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(A) $V_A = V_B = V_C$

29. The electric potential at a point in an electric field is the work done in bringing a unit positive charge from infinity to this point. If the potential difference between any two points at a surface is zero, the surface is called an equipotential surface. The shape of an equipotential surface can give direction of electric field at a point on it. It may be a closed surface or an open surface, depending on the charges creating the electric field.

(i) The shape of equipotential surface due to an isolated point charge is :

1

- (A) ellipsoidal, with charge at its one foci
- (B) plane surface not passing through point charge
- (C) spherical with charge at its centre
- (D) cylindrical with charge at its axis

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(C) Spherical with charge at its centre

(ii) The equipotential surfaces in a uniform electric field acting along + Z direction are :

1

- (A) planes parallel to the XY plane
- (B) concentric spherical surfaces
- (C) planes parallel to the YZ plane
- (D) planes parallel to the XZ plane

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(A) Planes parallel to the XY plane

(iii) Which of the following statements is *not* true for equipotential surfaces ?

1

- (A) Potentials at two equipotential surfaces are different.
- (B) No work is done in moving a charge on an equipotential surface.
- (C) Equipotential surfaces always enclose some volume.
- (D) Two equipotential surfaces do not intersect each other.

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(C) Equipotential surfaces always enclose some volume

- (iv) (a) The angle between the electric field at a point and outward normal at that point on an equipotential surface is : *1*
- (A) 90° (B) 45°
(C) 30° (D) 0°

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(A) 90°

(B) 45°

(C) 30°

(D) 0°

(D) 0^0

(iv) (b) A and B are two equipotential surfaces around a point charge Q. Surface A is closer to the charge than surface B. If a test charge is released between them, it will : 1

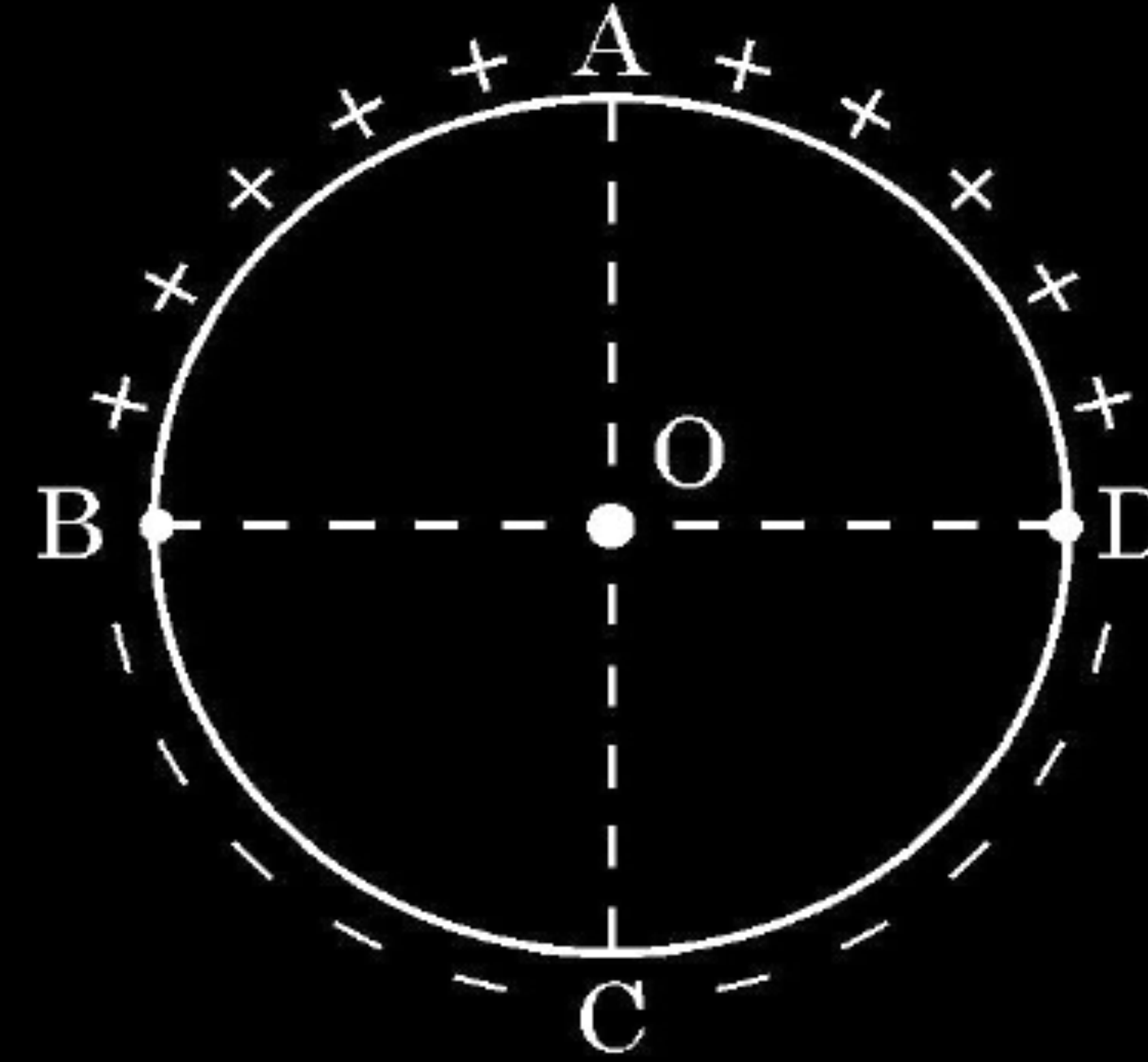
- (A) remain stationary
- (B) move from A to B
- (C) move from B to A
- (D) move around Q in a circular path between the equipotential surfaces A and B

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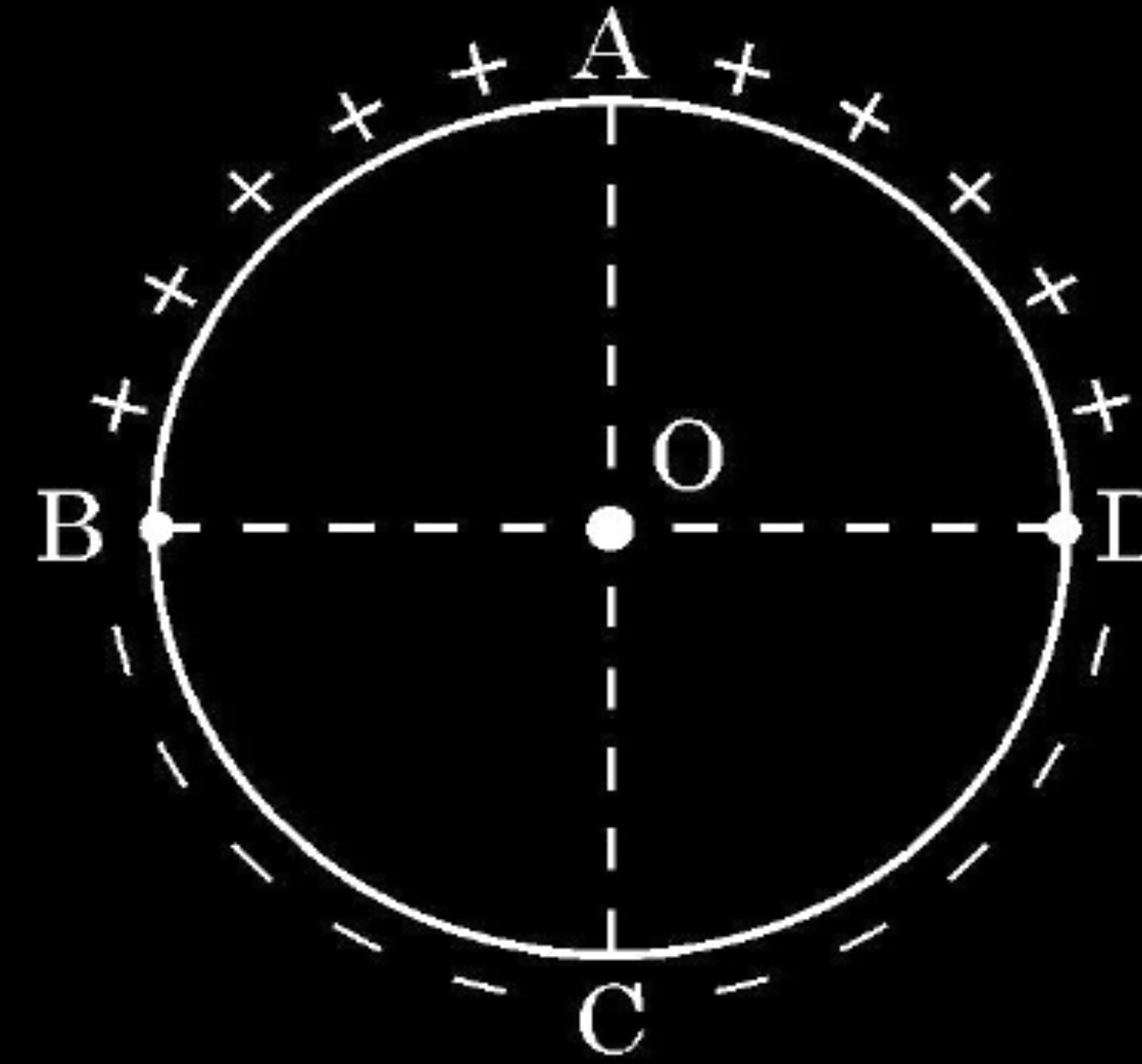
15. **Assertion (A)** : Equal amount of positive and negative charges are distributed uniformly on two halves of a thin circular ring as shown in figure. The resultant electric field at the centre O of the ring is along OC.

Reason (R) : It is so because the net potential at O is not zero.



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(C) Assertion (A) is true but Reason (R) is false.

23. Two conducting spherical shells A and B of radii R and $2R$ are kept far apart and charged to the same charge density σ . They are connected by a wire. Obtain an expression for final potential of shell A.

3

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Charge on spherical Shell A

$$q_A = 4\pi R^2 \sigma$$

Charge on speherical shell B

$$q_B = 4\pi (2R)^2 \sigma = 16\pi R^2 \sigma$$

After connecting by a wire , their potentials will become equal after sharing of charge.

$$\frac{1}{4\pi\epsilon_0} \frac{q'_A}{R} = \frac{1}{4\pi\epsilon_0} \frac{q'_B}{2R}$$

$$q'_B = 2q'_A$$

From conservation of charge

$$q_A + q_B = q'_A + q'_B$$

$$4\pi R^2 \sigma + 16\pi R^2 \sigma = 3q'_A$$

$$q'_A = \frac{20\pi R^2 \sigma}{3}$$

Final potential of Shell A

$$V_A = \frac{1}{4\pi\epsilon_0} \frac{q'_A}{R}$$

$$V_A = \frac{1}{4\pi\epsilon_0} \frac{20\pi R^2 \sigma}{3R}$$

$$V_A = \frac{5\sigma R}{3\epsilon_0}$$

$$\frac{1}{2}$$

$$\frac{1}{6}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

Alternatively

Charge on spherical shell A

$$q_A = 4\pi R^2 \sigma$$

Charge on spherical shell B

$$q_B = 4\pi (2R)^2 \sigma = 16\pi R^2 \sigma$$

After connecting by a wire , their potential will become equal after sharing of charges

Therefore the potential of shell A

$$\begin{aligned} V_A = V_{\text{common}} &= \frac{q_A + q_B}{C_A + C_B} \\ &= \frac{4\pi R^2 \sigma + 16\pi R^2 \sigma}{4\pi\epsilon_0 R + 4\pi\epsilon_0 (2R)} \\ &= \frac{5\sigma R}{3\epsilon_0} \end{aligned}$$

23. A thin spherical conducting shell of radius R has a charge q . A point charge Q is placed at the centre of the shell. Find (i) The charge density on the outer surface of the shell and (ii) the potential at a distance of $(R/2)$ from the centre of the shell.

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3

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- (i) When a point charge Q is placed at the centre of the shell, a charge $(-Q)$ is induced at its inner surface, consequently a net charge on outer surface of the shell $= q + Q$

Charge density on outer surface of the shell

$$\sigma = \frac{\text{charge}}{\text{Area}}$$

$$= \frac{q + Q}{4\pi R^2}$$

- (ii) Potential due to shell at a distance of $(R/2)$ from the centre of the shell

$$V_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

Potential due to charge Q at a distance of $(R/2)$ from the centre of the shell

$$V_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R/2}$$

Net potential at a distance of $(R/2)$ from the centre of the shell

$$V = V_1 + V_2$$

$$V = \frac{1}{4\pi\epsilon_0 R} (q + 2Q)$$

1. Consider a group of charges $q_1, q_2, q_3 \dots$ such that $\Sigma q \neq 0$. Then equipotentials at a large distance, due to this group are approximately :

(A) Plane

(B) Spherical surface

(C) Paraboloidal surface

(D) Ellipsoidal surface

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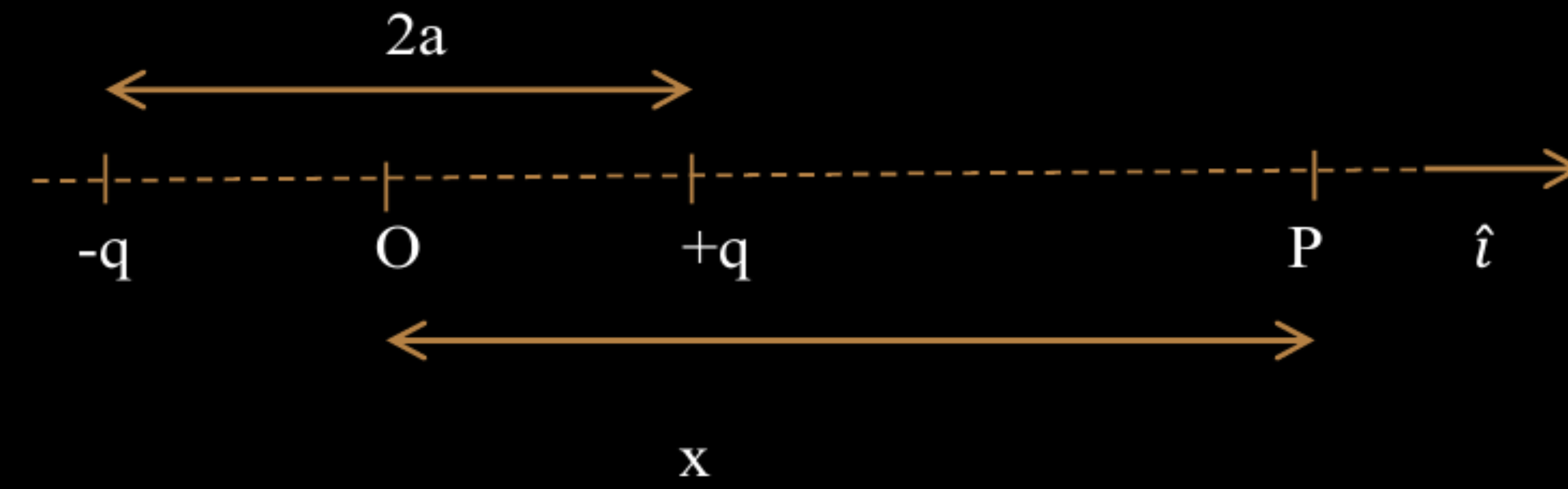
(D) Ellipsoidal surface

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(B) Spherical surface

- (b) (i) An electric dipole (dipole moment $\vec{p} = p \hat{i}$), consisting of charges $-q$ and q , separated by distance $2a$, is placed along the x -axis, with its centre at the origin. Show that the potential V , due to this dipole, at a point x , ($x \gg a$) is equal to $\frac{1}{4\pi\epsilon_0} \cdot \frac{\vec{p} \cdot \hat{i}}{x^2}$.
- (ii) Two isolated metallic spheres S_1 and S_2 of radii 1 cm and 3 cm respectively are charged such that both have the same charge density $\left(\frac{2}{\pi} \times 10^{-9}\right) \text{ C/m}^2$. They are placed far away from each other and connected by a thin wire. Calculate the new charge on sphere S_1 .

(i)



$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$V = V_{+q} - V_{-q}$$

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{(x-a)} - \frac{q}{(x+a)} \right]$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{x+a-x-a}{(x^2-a^2)} \right]$$

$$V = \frac{q}{4\pi\epsilon_0} \frac{2a}{(x^2-a^2)} = \frac{p}{4\pi\epsilon_0(x^2-a^2)}$$

As p is along x-axis, so

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{i}}{(x^2-a^2)}$$

If $x \gg a$

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{i}}{x^2}$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

(ii)

Charge on sphere S₁ :

Q₁ = surface charge density × surface Area

= $\left(\frac{2}{\pi} \times 10^{-9}\right) \times 4\pi (1 \times 10^{-2})^2$

= $8 \times 10^{-13} \text{ C}$

Charge on sphere S₂ :

Q₂ = surface charge density × surface Area

= $\left(\frac{2}{\pi} \times 10^{-9}\right) \times 4\pi (3 \times 10^{-2})^2$

= $72 \times 10^{-13} \text{ C}$

When connected by a thin wire they acquire a common potential V and the charge remains conserved.

$Q_1 + Q_2 = Q'_1 + Q'_2$

= $C_1V + C_2V$

$Q_1 + Q_2 = (C_1 + C_2)V$

Common potential(V) = $\frac{Q_1 + Q_2}{C_1 + C_2}$

½

½

½

$$C_1 = 4\pi\epsilon_0 r_1 = \frac{1}{9 \times 10^9} \times 10^{-2} = \frac{1}{9} \times 10^{-11} F$$

$$C_2 = 4\pi\epsilon_0 r_2 = \frac{1}{9 \times 10^9} \times 3 \times 10^{-2} = \frac{1}{3} \times 10^{-11} F$$

$$V = \frac{80 \times 10^{-13}}{\left(\frac{1}{9} + \frac{1}{3}\right) \times 10^{-11}} = 1.8 V$$

$$Q'_1 = C_1 V = \frac{1}{9} \times 10^{-11} \times 1.8$$

$$Q'_1 = 2 \times 10^{-12} C$$

½

½

17. *Assertion (A) :* The electrostatic field $\vec{E}$ is a conservative field.

Reason (R) : Line integral of electric field $\vec{E}$ around a closed path is non zero.

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Reason (R) : Line integral of electric field $\vec{E}$ around a closed path is non zero.

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(c) Assertion (A) is true, but Reason (R) is false.

- 24.** (a) A uniform electric field E of 500 N/C is directed along $+x$ axis. O, B and A are three points in the field having x and y coordinates (in cm) (0, 0), (4, 0) and (0, 3) respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B.

2

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2

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(a) (i) $V_{OA} = E (x_2 - x_1)$ $V_{OA} = 500 \times 0 = 0$ volt	$\frac{1}{2}$ $\frac{1}{2}$
(ii) $V_{OB} = -E (x_2 - x_1)$ $V_{OB} = -500 \times (4 \times 10^{-2})$ $= -20$ V	$\frac{1}{2}$ $\frac{1}{2}$

- (b) Two small conducting balls A and B of radius r_1 and r_2 have charges q_1 and q_2 respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium.

2

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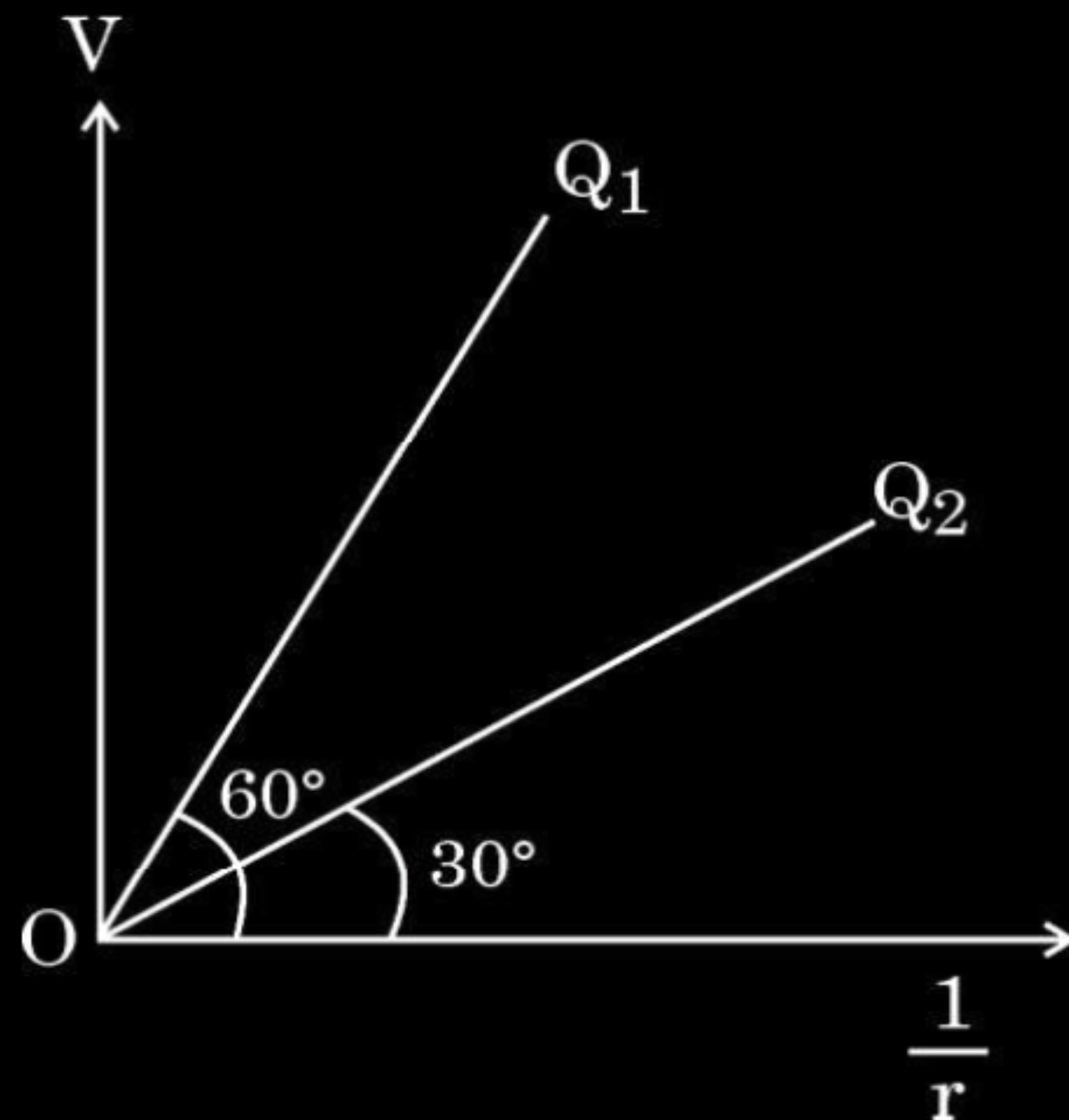
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According to law of conservation of charge		
$q_i = q_f$	$\frac{1}{2}$	CBSE 23
$q_1 + q_2 = q_1' + q_2' = Q$		
When two balls are connected with wire		
$V_1 = V_2$	$\frac{1}{2}$	
$\frac{kq_1'}{r_1} = \frac{kq_2'}{r_2} \text{ or } \frac{q_1'}{r_1} = \frac{q_2'}{r_2}$		
$q_1' r_2 = q_2' r_1$		
$q_1' r_2 = (Q - q_1') r_1$		
$q_1' r_2 = Q r_1 - q_1' r_1$		
$q_1' (r_1 + r_2) = Q r_1$		
$q_1' = \frac{Q r_1}{r_1 + r_2} = \frac{(q_1 + q_2) r_1}{r_1 + r_2}$	$\frac{1}{2}$	
$q_2' = Q - q_1'$		
$= Q - \frac{Q r_1}{r_1 + r_2}$		
$= \frac{Q r_2}{r_1 + r_2} = \frac{(q_1 + q_2) r_2}{r_1 + r_2}$	$\frac{1}{2}$	

- 34.** Electrostatics deals with the study of forces, fields and potentials arising from static charges. Force and electric field, due to a point charge is basically determined by Coulomb's law. For symmetric charge configurations, Gauss's law, which is also based on Coulomb's law, helps us to find the electric field. A charge/a system of charges like a dipole experience a force/torque in an electric field. Work is required to be done to provide a specific orientation to a dipole with respect to an electric field.

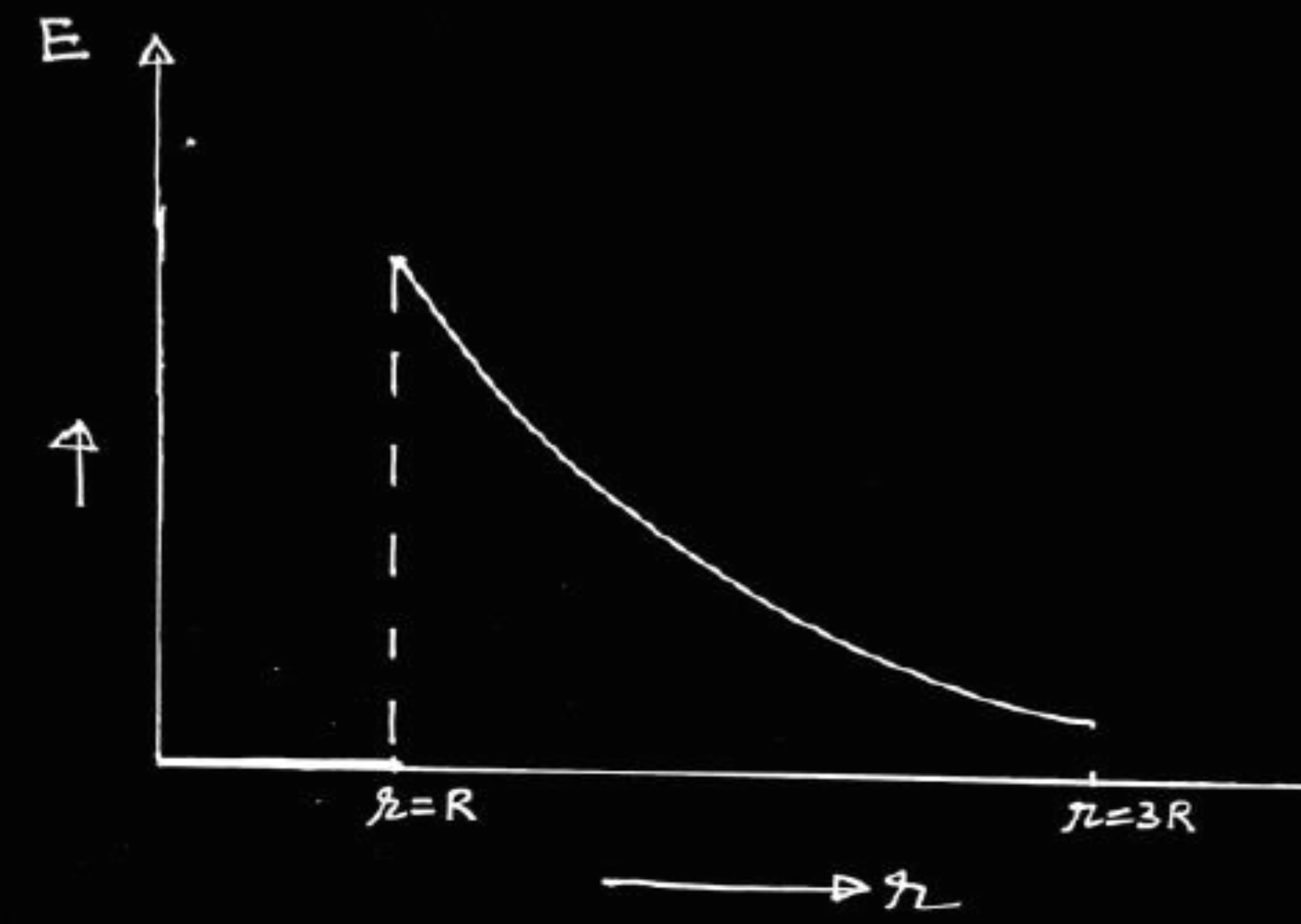
CBSE 23

Answer the following questions based on the above :



- (a) Consider a uniformly charged thin conducting shell of radius R . Plot a graph showing the variation of $|\vec{E}|$ with distance r from the centre, for points $0 \leq r \leq 3R$. 1
- (b) The figure shows the variation of potential V with $\frac{1}{r}$ for two point charges Q_1 and Q_2 , where V is the potential at a distance r due to a point charge. Find $\frac{Q_1}{Q_2}$. 1

(a)



1

(b) $\because V = k Q/r$

Slope of graph is proportional to Q

$$\frac{Q_1}{Q_2} = \frac{\tan 60^\circ}{\tan 30^\circ} = 3$$

 $\frac{1}{2}$ $\frac{1}{2}$

(c) $U = -p E \cos \theta$

$$\theta = 0^\circ$$

$$U = - (6 \times 10^{-7}) \times (10^4)$$

$$U = -6 \times 10^{-3} \text{ J}$$

 $\frac{1}{2}$ $\frac{1}{2}$

1

OR

$$\because \text{Work done } W = -p E (\cos \theta_2 - \cos \theta_1)$$

$$\text{where } \theta_2 = 180^\circ, \theta_1 = 90^\circ$$

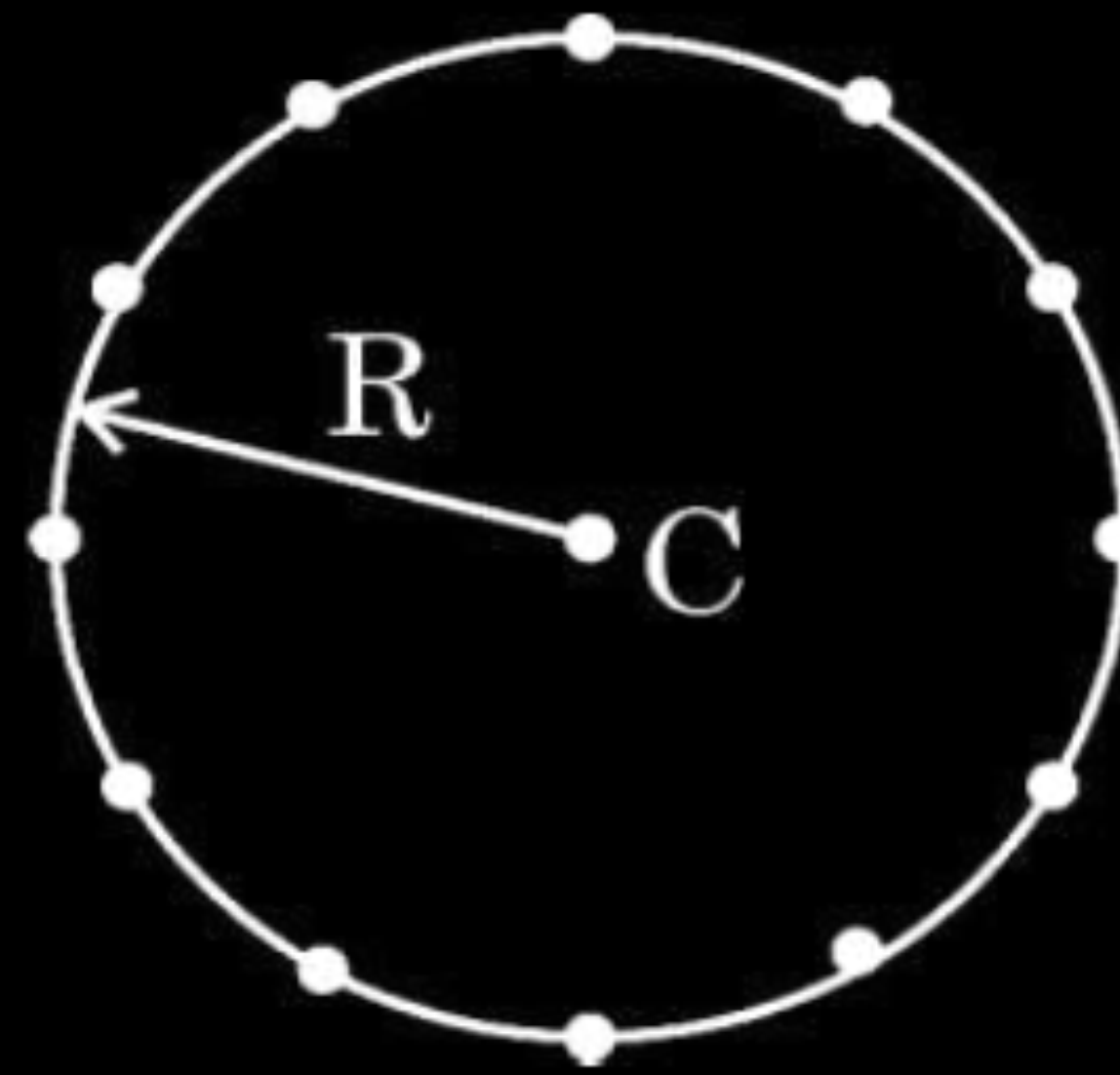
$$\Rightarrow W = -p E (\cos 180^\circ - \cos 90^\circ)$$

$$W = +p E$$

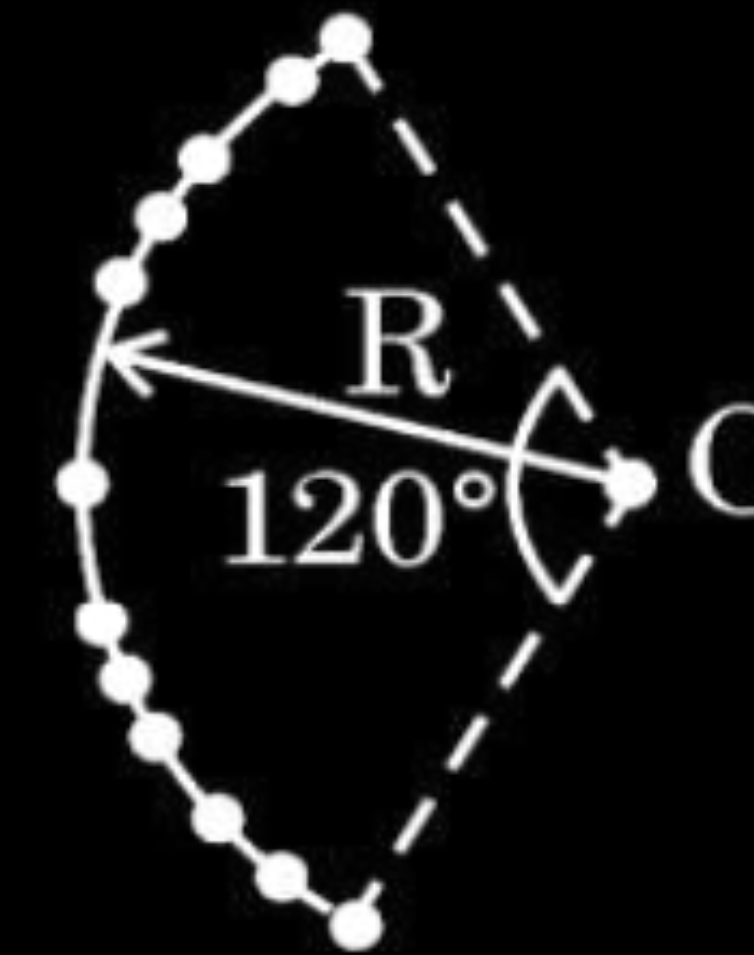
 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

- 26.** (a) Twelve negative charges of same magnitude are equally spaced and fixed on the circumference of a circle of radius R as shown in Fig. (i). Relative to potential being zero at infinity, find the electric potential and electric field at the centre C of the circle.
- (b) If the charges are unequally spaced and fixed on an arc of 120° of radius R as shown in Fig. (ii), find electric potential at the centre C . 3

CBSE 23



(i)



(ii)

(a)

Electric potential due to point charge

$$V = \frac{kq}{R}$$

Value of each charge = $-q$, Total charge = $-12q$

$$\text{Total potential } V = \frac{k(-12q)}{R}$$

$$V = \frac{-12kq}{R} = \frac{-12q}{4\pi\epsilon_0 R}$$

By symmetry the resultant of all electric field vectors becomes zero.

So electric field is zero.

(b)

Electric potential is a scalar quantity and does not depend on placement of charges

$$\text{Therefore } V = \frac{-12kq}{R} = \frac{-12q}{4\pi\epsilon_0 R}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

1

18. **Assertion (A)** : Work done in moving a charge around a closed path, in an electric field is always zero.

Reason (R) : Electrostatic force is a conservative force.

CBSE 23

1

18. **Assertion (A)** : Work done in moving a charge around a closed path, in an electric field is always zero.

Reason (R) : Electrostatic force is a conservative force.

CBSE 23

1

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

28. (a) Two charged conducting spheres of radii a and b are connected to each other by a wire. Find the ratio of the electric fields at their surfaces.

CBSE 23

28. (a) Two charged conducting spheres of radii a and b are connected to each other by a wire. Find the ratio of the electric fields at their surfaces.

CBSE 23

3

When connected by a conducting wire both spheres will be at the same potential.

$$\therefore k \frac{q_1}{a} = k \frac{q_2}{b}$$

$$\therefore \frac{q_1}{q_2} = \frac{a}{b}$$

$$\frac{E_1}{E_2} = \frac{k \frac{q_1}{a^2}}{k \frac{q_2}{b^2}}$$

$$\frac{E_1}{E_2} = \frac{b}{a}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

2. A point P lies at a distance x from the mid point of an electric dipole on its axis. The electric potential at point P is proportional to

1

(a) $\frac{1}{x^2}$

(b) $\frac{1}{x^3}$

(c) $\frac{1}{x^4}$

(d) $\frac{1}{x^{1/2}}$

CBSE 23

2. A point P lies at a distance x from the mid point of an electric dipole on its axis. The electric potential at point P is proportional to **1**

(a) $\frac{1}{x^2}$

(b) $\frac{1}{x^3}$

(c) $\frac{1}{x^4}$

(d) $\frac{1}{x^{1/2}}$

CBSE 23

(a)
 $\frac{1}{x^2}$

37. (a) Write two important characteristics of equipotential surfaces.
- (b) The magnitude of electric field (in NC^{-1}) in a region varies with the distance r (in m) as

$$E = 10r + 5$$

CBSE 20

By how much does the electric potential increase in moving from point at $r = 1$ m to a point at $r = 10$ m.

37. (a) Write two important characteristics of equipotential surfaces.
- (b) The magnitude of electric field (in NC^{-1}) in a region varies with the distance r (in m) as

$$E = 10r + 5$$

CBSE 20

By how much does the electric potential increase in moving from point at $r = 1$ m to a point at $r = 10$ m.

(a) For equipotential surfaces	
(i) Potential has the same value at all points on the surface.	
(ii) Electric field is normal to the equipotential surface at all points	1+1
(iii) Work done in moving any charge between any two points on the equipotential surface is zero (any two)	

(b)	
$E = 10r + 5$	
$dV = -E \cdot dr$	
$\int dV = - \int_1^{10} \vec{E} \cdot d\vec{r}$	$\frac{1}{2}$
$= - \int_1^{10} (10r + 5) dr$	
$V = - \left[\int_1^{10} 10r \, dr + \int_1^{10} 5 \, dr \right]$	$\frac{1}{2}$
$V = 10 \left[\frac{r^2}{2} \right]_1^{10} + 5(r)_1^{10}$	
$V = -5[100 - 1] + 5[10 - 1]$	$\frac{1}{2}$
$V = -5 \times 99 + 5 \times 9 = -540V$	

12. A proton released from rest in an electric field, will start moving towards a region of _____ potential in the field.

CBSE 20

12. A proton released from rest in an electric field, will start moving towards a region of _____ potential in the field.

1

CBSE 20

Decreasing/Lower

16. The work done in moving a charge particle between two points in an uniform electric field, does not depend on the path followed by the particle. Why ?

CBSE 20

16. The work done in moving a charge particle between two points in an uniform electric field, does not depend on the path followed by the particle. Why ?

CBSE 20

1

Because the electrostatic force is conservative in nature

Alternatively:-

Electric field is conservative in nature / work done by or against the electric field does not depend upon the path followed.

1

1. If a positive charge is displaced against the electric field in which it was situated, then

1

- (A) work will be done by the electric field on the charge.
- (B) the intensity of the electric field decreases.
- (C) energy of the system will decrease.
- (D) energy will be provided by external source displacing the charge.

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CBSE 20

(D)

energy will be provided by external source displacing the charge.

Why is the electrostatic potential inside a charged conducting shell constant throughout the volume of the conductor ?

CBSE 19

Why is the electrostatic potential inside a charged conducting shell constant throughout the volume of the conductor ?

CBSE 19

$E = 0$ inside the conductor & has no tangential component on the surface.

$\therefore$ No work is done in moving charge inside or on the surface of the conductor.

& Potential is constant.

[Even if a student writes “because $E = 0$ inside the conductor” - award full marks Or No work is done in moving a charge inside the conductor - award full marks.]

$\frac{1}{2}$

$\frac{1}{2}$

A charge Q is distributed over the surfaces of two concentric hollow spheres of radii r and R ($R \gg r$), such that their surface charge densities are equal. Derive the expression for the potential at the common centre.

A charge Q is distributed over the surfaces of two concentric hollow spheres of radii r and R ($R \gg r$), such that their surface charge densities are equal. Derive the expression for the potential at the common centre.

$$Q = q_1 + q_2$$

$$= 4\pi\sigma(r^2 + R^2)$$

Potential at common centre

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r} + \frac{q_2}{R} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \times \left[\frac{4\pi r^2 \sigma}{r} + \frac{4\pi R^2 \sigma}{R} \right]$$

$$= \frac{(r + R)\sigma}{\epsilon_0}$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{Q(r + R)}{r^2 + R^2} \right]$$

ORF 10

$\frac{1}{2}$

$\frac{1}{2}$

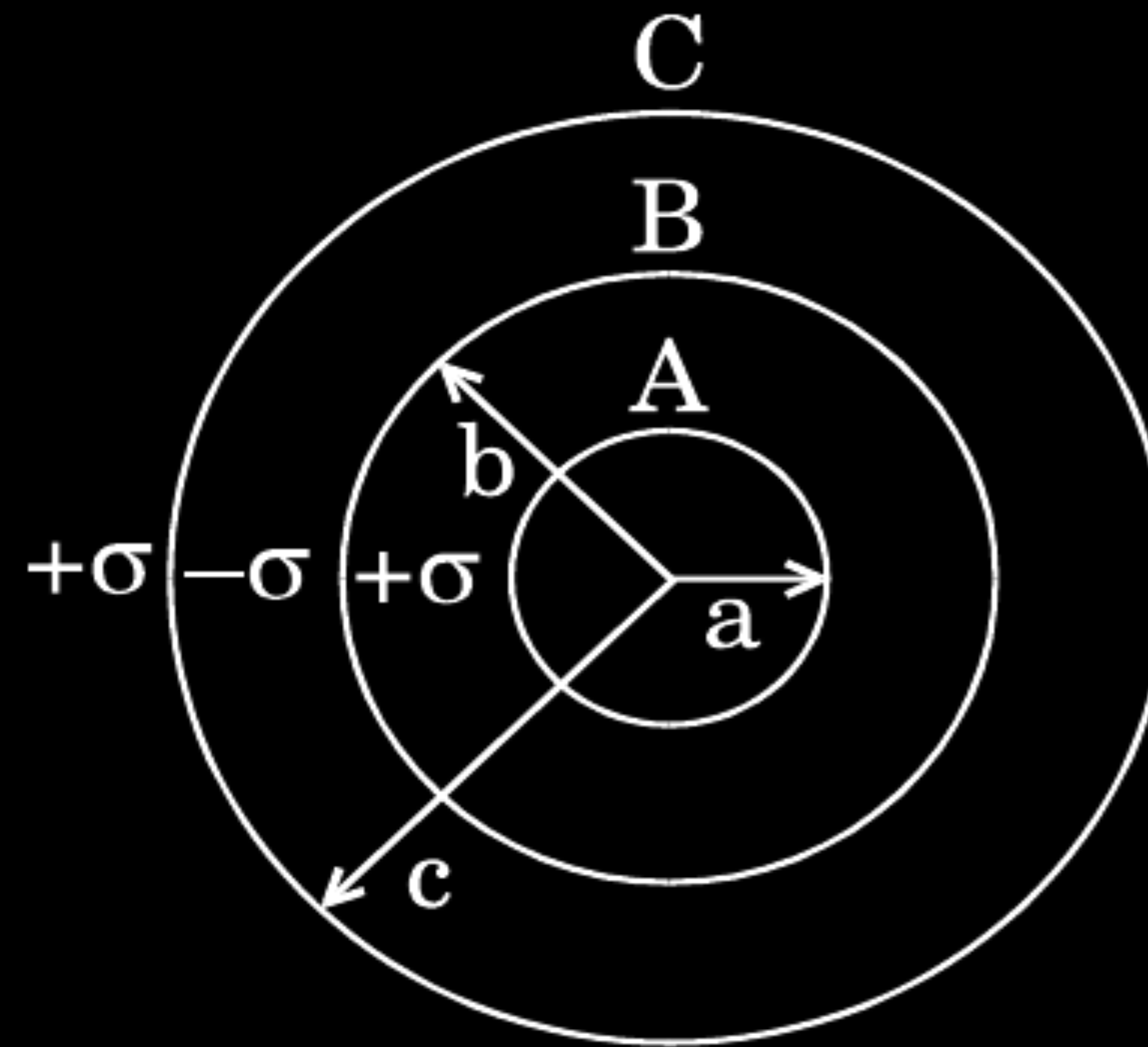
$\frac{1}{2}$

$\frac{1}{2}$

Three concentric metallic shells A, B and C of radii a , b and c ($a < b < c$) have surface charge densities $+\sigma$, $-\sigma$ and $+\sigma$ respectively as shown.

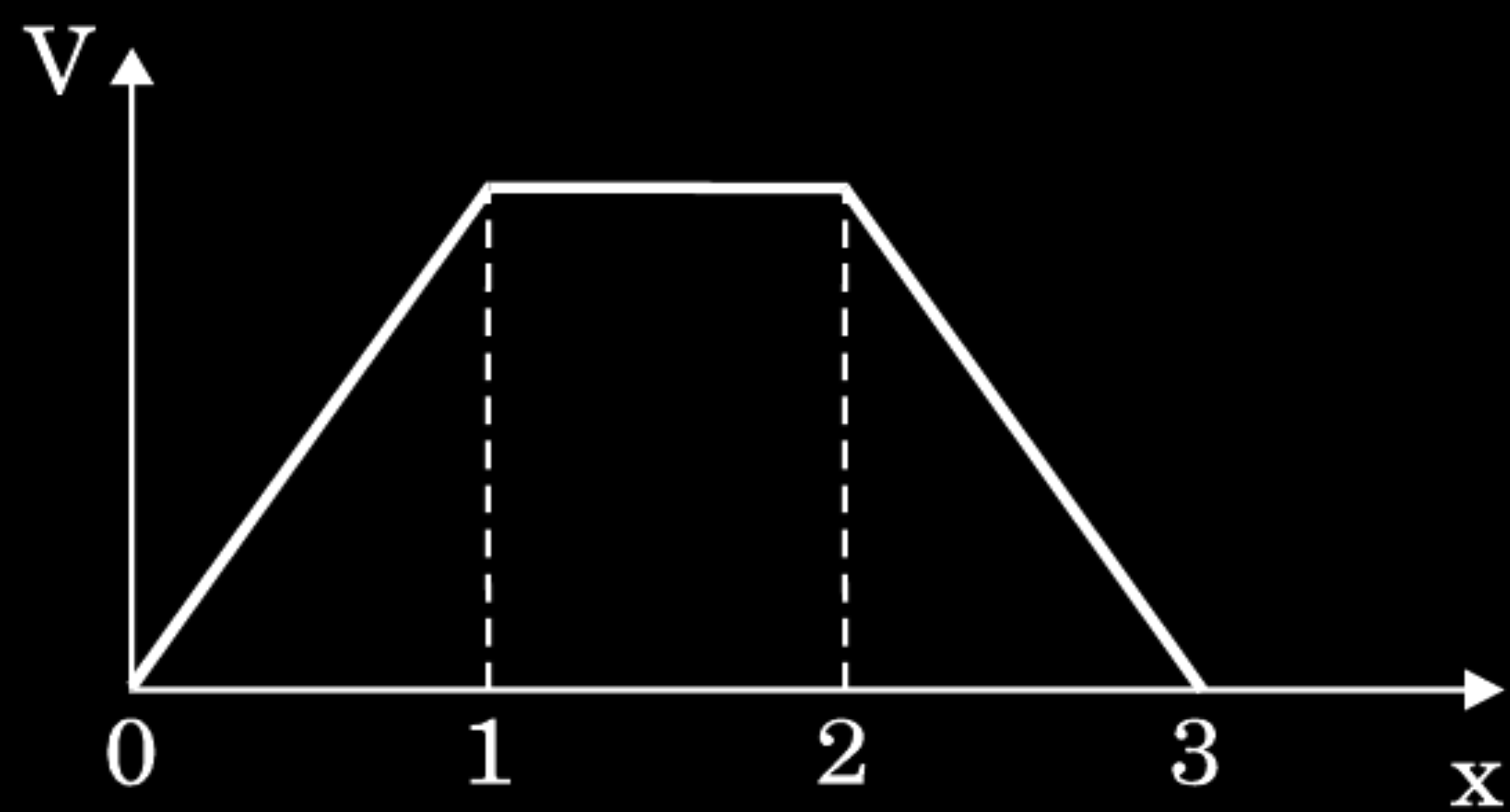
- (a) Obtain the expressions for the potential of three shells A, B and C.
- (b) If shells A and C are at the same potential, obtain the relation between a , b and c .

CBSE 19

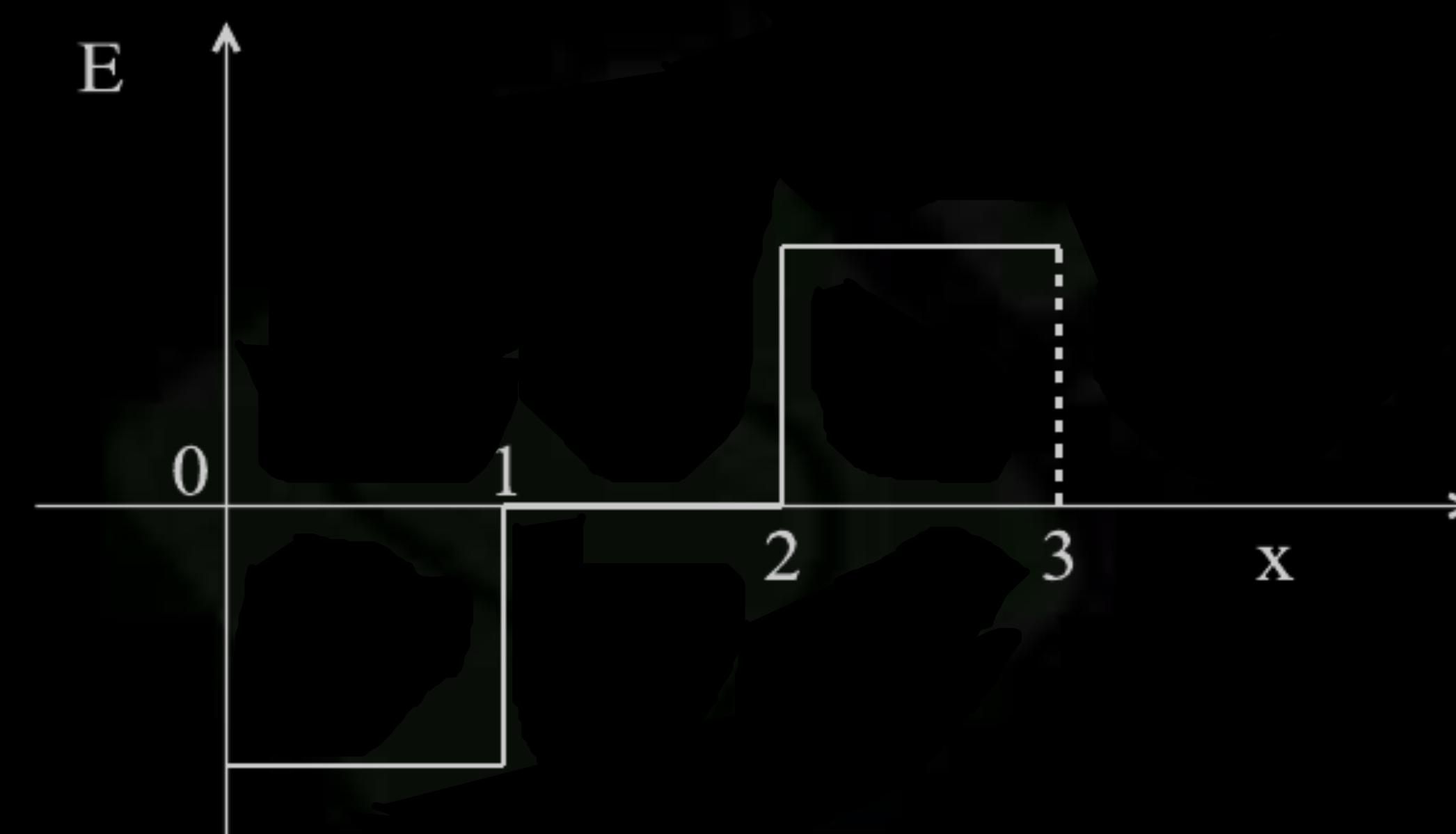


$V_A = \frac{kQ_A}{a} + \frac{kQ_B}{b} + \frac{kQ_C}{c}$	
$V_A = \frac{1}{4\pi\epsilon_0} \left[\frac{4\pi a^2 \sigma}{a} - \frac{4\pi b^2 \sigma}{b} + \frac{4\pi c^2 \sigma}{c} \right]$	$\frac{1}{2}$
$= \frac{\sigma}{\epsilon_0} [a - b + c]$	$\frac{1}{2}$
$V_B = \frac{kQ_A}{b} + \frac{kQ_B}{b} + \frac{kQ_C}{c}$	
$V_B = \frac{1}{4\pi\epsilon_0} \left[\frac{4\pi a^2 \sigma}{b} - \frac{4\pi b^2 \sigma}{b} + \frac{4\pi c^2 \sigma}{c} \right]$	$\frac{1}{2}$
$V_B = \frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{b} + c \right]$	$\frac{1}{2}$
$V_C = \frac{kQ_A}{c} + \frac{kQ_B}{c} + \frac{kQ_C}{c}$	
$V_C = \frac{1}{4\pi\epsilon_0} \left[\frac{4\pi a^2 \sigma}{c} - \frac{4\pi b^2 \sigma}{c} + \frac{4\pi c^2 \sigma}{c} \right]$	$\frac{1}{2}$
$= \frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2 + c^2}{c} \right]$	
$V_A = V_C$	
$a - b + c = \frac{a^2 - b^2}{c} + c$	$\frac{1}{2}$
$c = a + b$	

- (c) The electric potential as a function of distance 'x' is shown in the figure. Draw a graph of the electric field E as a function of x .



iii)



Draw equipotential surfaces for an electric dipole.

Draw equipotential surfaces for an electric dipole.

CBSE 19



(Even if a student mentions or draws equatorial plane, award 1 mark.)

1

A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge Q .

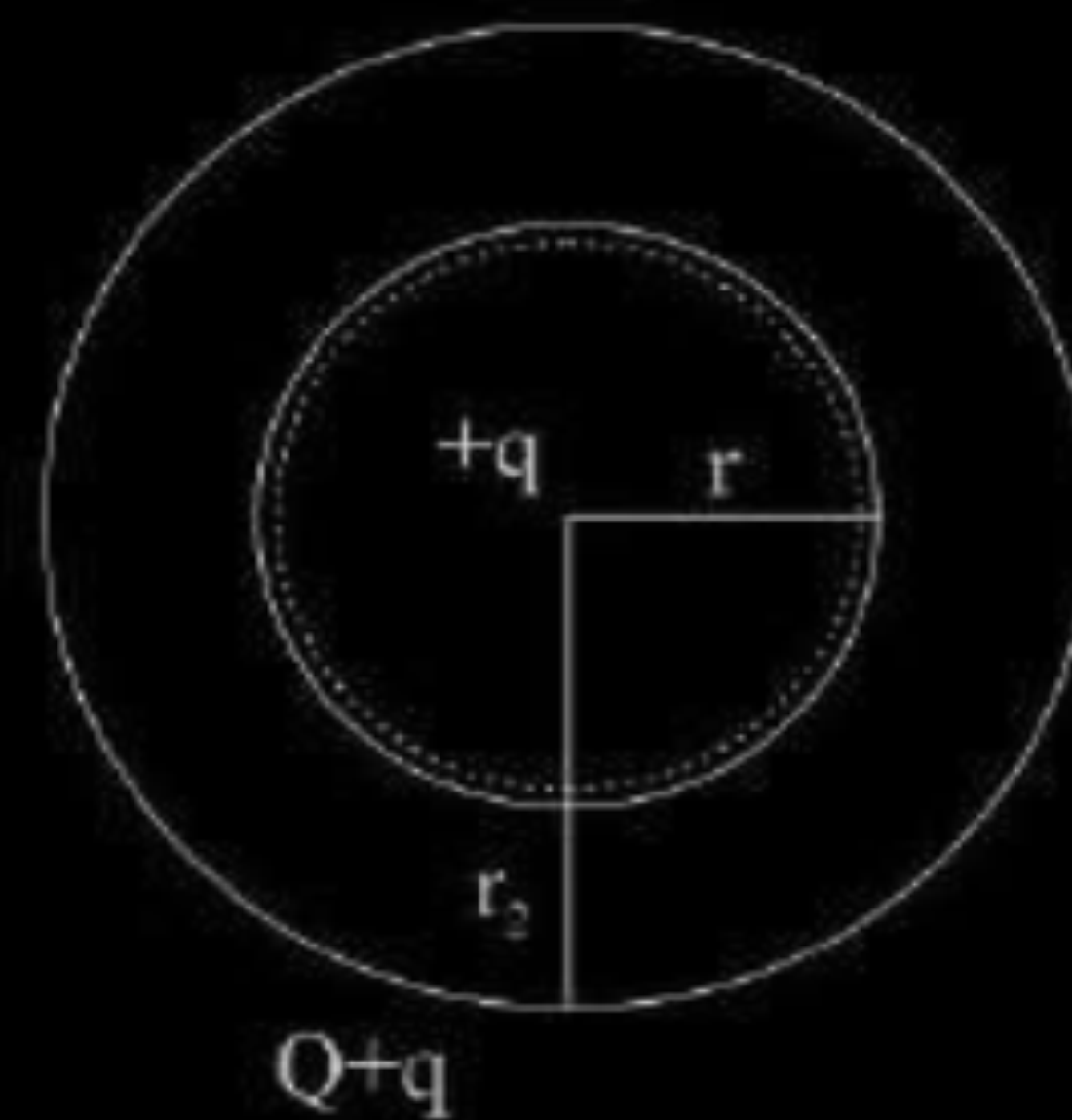
- (a) A charge q is placed at the centre of the shell. Find out the surface charge density on the inner and outer surfaces of the shell.
- (b) Is the electric field inside a cavity (with no charge) zero; independent of the fact whether the shell is spherical or not ? Explain.

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CBSE 19

(a) Diagram



The surface charge density on inner surface of the shell is $\sigma_1 = -\frac{q}{4\pi r_1^2}$

The surface charge density on outer shell is $\sigma_2 = \frac{Q+q}{4\pi r_2^2}$

(b) Consider a Gaussian surface inside the shell, net flux is zero since $q_{net} = 0$. According to Gauss's law it is independent of shape and size of shell.

1/2

1

1/2

1

A point charge $+Q$ is placed at point O as shown in the figure. Is the potential difference $V_A - V_B$ positive, negative or zero ?



CBSE 16

A point charge $+Q$ is placed at point O as shown in the figure. Is the potential difference $V_A - V_B$ positive, negative or zero ?



CBSE 16

Positive

What is the amount of work done in moving a point charge Q around a circular arc of radius ' r ' at the centre of which another point charge ' q ' is located ?

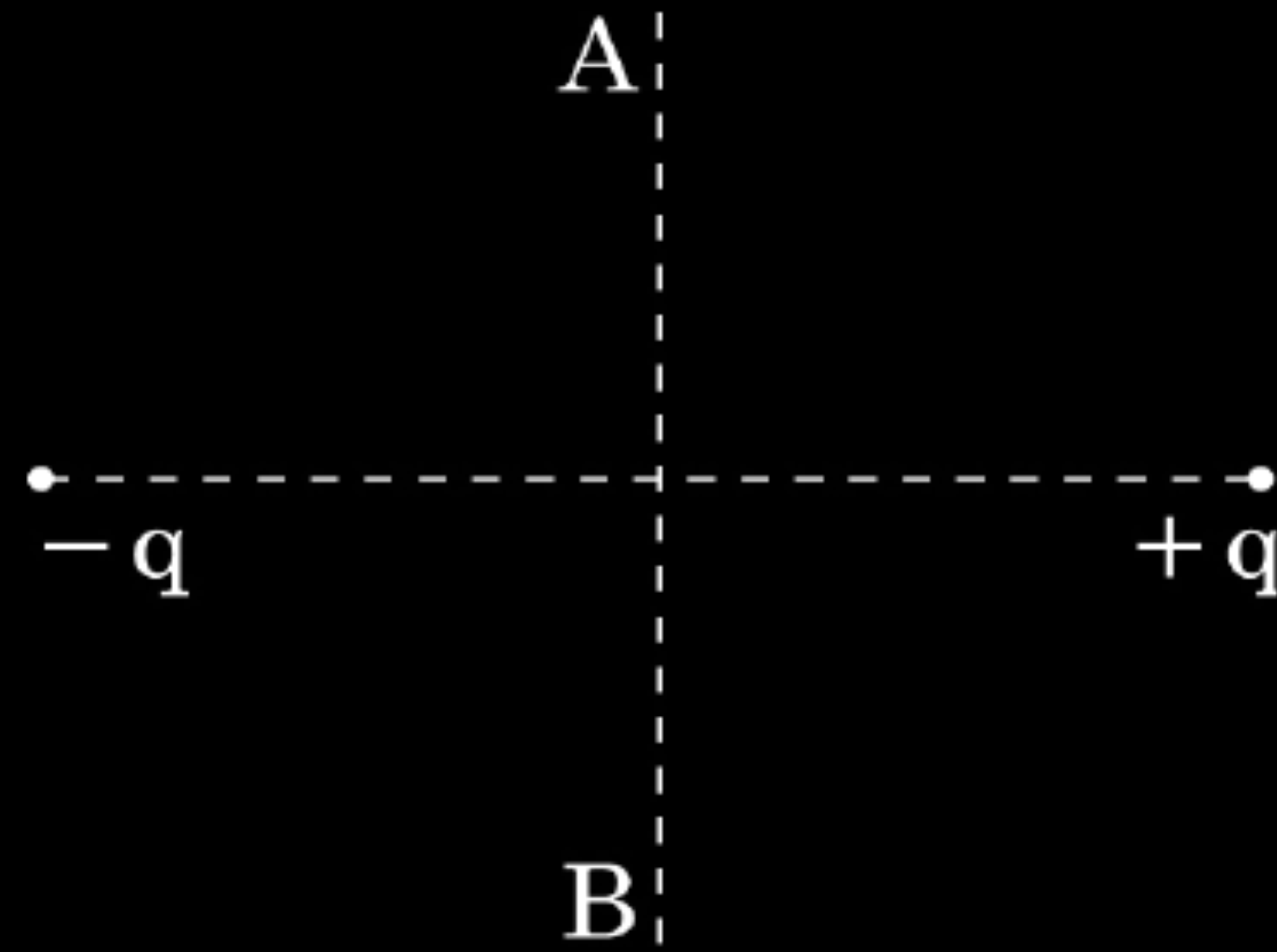
CBSE 16

What is the amount of work done in moving a point charge Q around a circular arc of radius ' r ' at the centre of which another point charge ' q ' is located ?

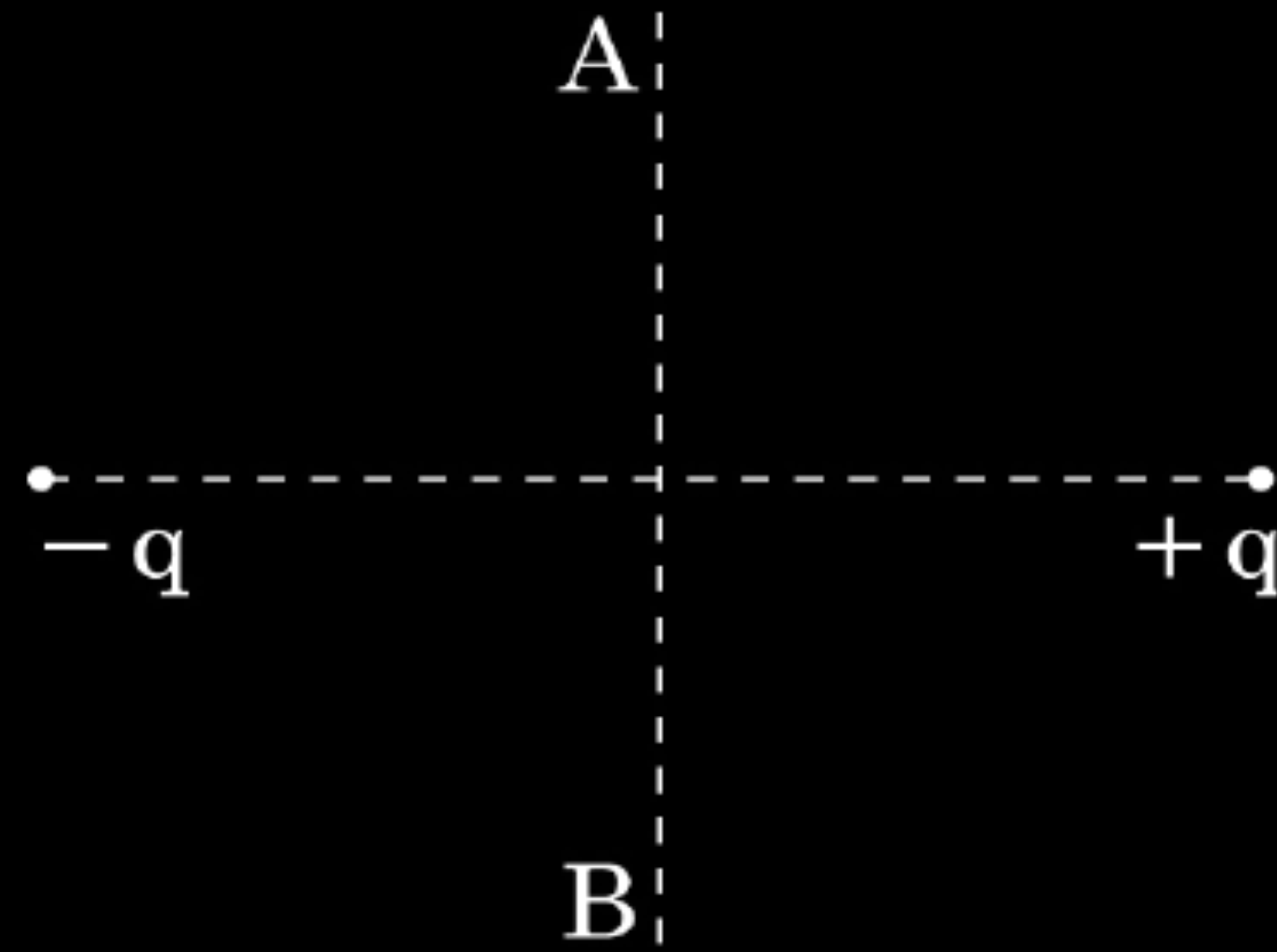
CBSE 16

Zero / No work done / None

A charge 'q' is moved from a point A above a dipole of dipole moment 'p' to a point B below the dipole in equatorial plane without acceleration. Find the work done in the process.



A charge 'q' is moved from a point A above a dipole of dipole moment 'p' to a point B below the dipole in equatorial plane without acceleration. Find the work done in the process.



CBSE 16

No work is done /

$$W = qV_{AB} = q \times 0 = 0$$

Define an equipotential surface. Draw equipotential surfaces :

- (i) in the case of a single point charge and
- (ii) in a constant electric field in Z-direction.

Why the equipotential surfaces about a single charge are not equidistant ?

- (iii) Can electric field exist tangential to an equipotential surface ? Give reason.

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Why the equipotential surfaces about a single charge are not equidistant ?

- (iii) Can electric field exist tangential to an equipotential surface ? Give reason.

Surface with a constant value of potential at all points on the surface.

ii. 

$V \propto \frac{1}{r}$

iii.No

If the field lines are tangential, work will be done in moving a charge on the surface which goes against the definition of equipotential surface.

1/2

1/2

1/2

1/2

